

GOV 51 - Final Project: Draft Analysis

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GitHub Repository Link: <https://github.com/daniel-e-cabrera/GOV-51-Final-Project>

INTRODUCTION

Research Question: For my final project, I am interested in answering the question: How does sexuality impact an individual's employment? More specifically, does sexual orientation predict the likelihood of experiencing workplace discrimination? And, if so, what is the strength of this

Motivation: As a gay man who is about to join the workforce, I have started to become concerned about whether my sexual orientation will impact my search for employment. As a Harvard student, I have often been told that finding decent employment is not a concern for students from top institutions; however, I question whether that is true for all Harvard students. I am particularly interested in whether, once I am in an interview and visible to an employer, my sexual orientation may influence how I am evaluated. Similarly, even if I am employed, what will my everyday experience be like? Will I be discriminated against because of my sexuality? Additionally, after having spoken to others at the college about my worries, I have realized that this isn't just an individualistic experience, but one that many others share. Therefore, it feels necessary to conduct a study to better understand whether and how sexual orientation influences employment for individuals like myself.

Study Type: The research question I am asking is predictive. In other words, it is focused on determining whether there is a relationship between variables such as sexuality and outcomes such as workplace experiences. Said differently, my final project examines how sexual orientation predicts the likelihood of experiencing employment discrimination. I am not attempting to establish that being non-heterosexual causes discrimination, but rather examining whether sexuality is associated with differences in reported workplace treatment. This project is not causal because I am not using an experimental or quasi-experimental design, and individuals are not randomly assigned a sexual orientation, which means I cannot isolate causal effects or rule out confounding variables. At the same time, this project is not purely descriptive, because I am not simply summarizing patterns in the data; instead, I am using statistical models to estimate the strength of the relationship between sexuality and employment discrimination and to evaluate how well sexuality predicts these outcomes, making this project prediction-based.

Dataset: In order to answer these research questions, I am using data from the **2016 General Social Survey (GSS)**. Within my specific data subset, set to 2016 (the only year where individuals were asked both about their sexuality and workplace discrimination), there are 42 individuals who identified as gay, lesbian, or bisexual (Independent Variable) out of 1,189 total respondents (who answered both the sexuality and workplace discrimination questions). Key variables I will examine include whether respondents report experiencing workplace discrimination over the past five years (possible responses include: Yes, No, Inapplicable, No answer, and Do not Know/Cannot Choose). Notably, this is self-reported work discrimination. Additionally, I will include control variables such as race to better isolate the association between sexuality and employment-related outcomes.

Additionally, I plan use the **Pew Research Center 2013 LGBT Survey** data set in order to compliment my findings from the previous GSS data set, due to a low number of respondents. In their surveying, Pew Research conducted an analysis of the experiences and attitudes of the LGBT population between April 11th and 29th of 2013. In their survey they gathered a sample of 1,197 self-identified LGBT adults, 18 years of age or older. The survey, conducted by the GfK Group using KnowledgePanel, its nationally representative online research panel, recruited through probability sampling methods and include both those with internet access and those without (KnowledgePanel provides internet access for those who do not have it and, if needed, a device to access the internet when they join the panel). A combination of random digit dialing and address-based sampling methodologies were used to recruit panel members. Furthermore, the survey was conducted in English. Panel members complete an annual profile survey that includes a range of demographic, attitudinal and behavioral questions. Additionally, in order to avoid any potential response bias related to the topic of the survey, the invitations did not describe the content or target population of the study. The Pew Research survey relies on a random sample of adults who have agreed to be a part of the GfK KnowledgePanel and complete surveys in exchange for a modest monetary reward. Interviews are conducted online, with no interviewer present. Notably, it is possible that some LGBT individuals who would be unwilling to disclose their status in other contexts are willing to identify themselves as LGBT in this panel. Key variables that I plan to utilize include x, y, and z.

Hypotheses: My formal hypotheses are as follows:

H0 (null): There is no difference in reported workplace discrimination between non-heterosexual and heterosexual individuals.

H1: Non-heterosexual individuals will report higher levels of workplace discrimination than heterosexual individuals.

H2: Sexual orientation will remain a significant predictor of workplace discrimination, even when controlling for other salient variables such as race and gender.

H3: The effect of sexual orientation on workplace discrimination will vary by gender, with non-heterosexual men reporting higher levels of discrimination than non-heterosexual women.

Literature Review: According to a [study](#) done by UCLA Law in 2022, “using survey data collected in May 2021 from 935 LGBT adults in the workforce” researchers found that about 75% of their 935 LGB adults said they experienced workplace discrimination (UCLA Law School Williams Institute, 2022). Interestingly, they also found that “higher rates of concealing LGB identity among bisexual employees may mask the extent to which bisexual employees experience unfair treatment at work” (Ibid). Notably, their study found that “cisgender bisexual and gay men often report higher rates of unfair treatment compared to lesbian and bisexual women” (Ibid). Overall, this study offers several key insights that inform the foundation of my research. Most importantly, it demonstrates that the non-heterosexual community should not be treated as a monolithic group, but rather as one with many nuances. It also highlights that levels of identity disclosure, particularly how “out” individuals are in the workplace, play a significant role in shaping reported experiences of discrimination.

Building on studies conducted by the [UCLA Law School Williams Institute](#), [Weistroffer et al.](#) and [Hoel et al.](#), this project seeks to move beyond the foundational insights these works provide. While existing research consistently finds that LGB individuals experience higher levels of workplace discrimination compared to their heterosexual counterparts, there remains important disagreement regarding how these experiences vary by gender. For instance, the UCLA Law School study finds that gay men report higher levels of workplace discrimination, whereas Hoel et al. report the inverse pattern. This divergence points to the need for further investigation.

Moreover, prior research has not fully quantified the relative impact of sexual orientation compared to other important social identities. Because workplace discrimination is shaped by multiple, intersecting factors, this study introduces a novel contribution by analyzing the predictive power of sexual orientation alongside other key variables, such as race and gender.

METHODS

By sub setting the GSS data set to only respondents with definitive yes/no answers, thus removing Inapplicable, No answer, and Do not Know/Cannot Choose responses, this introduces the possibility of selection bias.

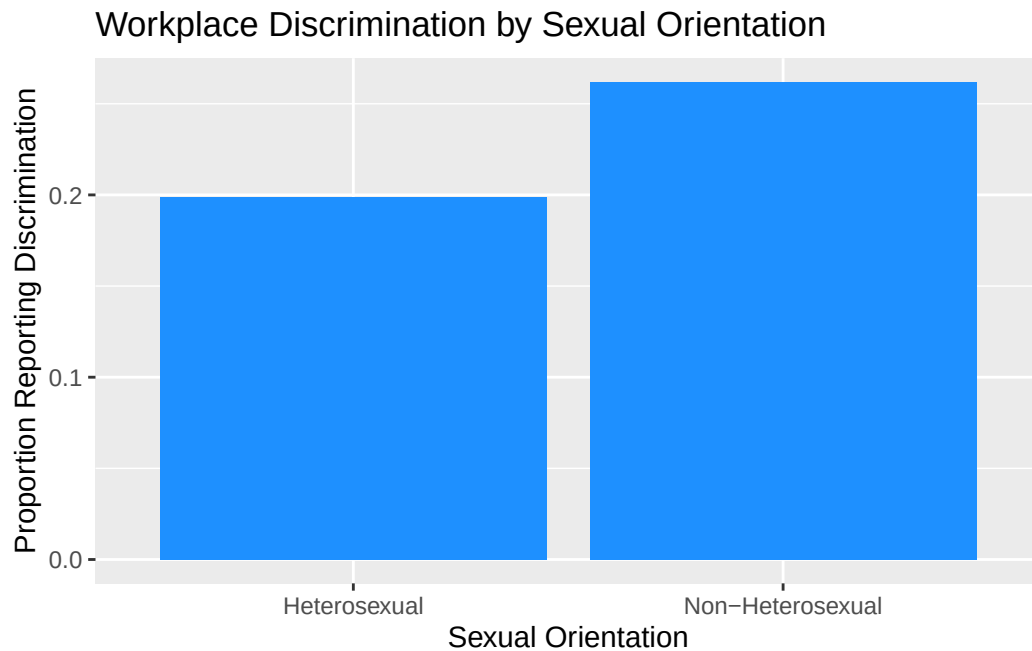
Preliminary GSS Data Visuals

Table 1: Discrimination Rate by Sexual Orientation

sexorient	Individuals	Discrimination_Rate
Heterosexual	1147	0.199
Non-Heterosexual	42	0.262

Here, is a simple 2 x 3 table that shows the number of heterosexual sexual individuals (1156) who answered both the questions about sexuality and workplace discrimination, and the number of non-heterosexual individuals (42), defined as answering: gay, lesbian or homosexual,

or bisexual. Additionally, for each group (heterosexual and non-heterosexual) we see their discrimination rates (or the mean of their answers to experiencing workplace discrimination over the past five years). For heterosexuals, the discrimination rate was 19.72% and for non-heterosexuals, it was 26.19%. Therefore, non-heterosexual respondents are approximately 6.5 percentage points more likely to report workplace discrimination.



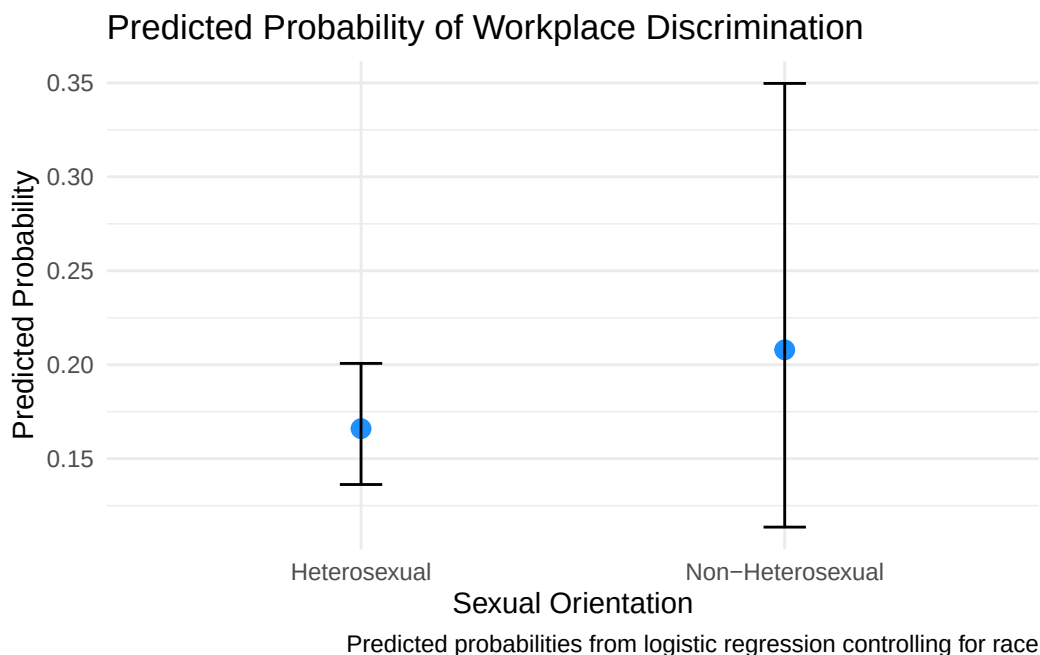
This simple bar chart, gives us a visual of the proportion of individuals who said yes to experiencing workplace discrimination over the past five years for each group. The results indicate a clear disparity between groups: individuals identifying as non-heterosexual report higher levels of workplace discrimination than their heterosexual counterparts.

This table shows the results of a logistic regression examining the relationship between sexual orientation and the likelihood of reporting workplace discrimination. The coefficient for non-heterosexual respondents is positive (0.368), indicating that individuals who identify as non-heterosexual have higher log-odds of reporting workplace discrimination compared to heterosexual respondents. However, the estimate is not statistically significant, as shown by the relatively large standard error and lack of significance stars, meaning that we cannot reject the null hypothesis of no effect. This is likely due to small sample size. Ideally, I will find another data set (that will be complementary to the GSS one) which has larger sample sizes.

Table 2: Logistic Regression: Workplace Discrimination on Sexual Orientation

	(1)
(Intercept)	−1.394*** (0.074)
sexorientNon-Heterosexual	0.358 (0.359)
Num.Obs.	1189
AIC	1196.3
BIC	1206.5
Log.Lik.	−596.167
RMSE	0.40

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001



The second figure presents predicted probabilities of reporting workplace discrimination by sexual orientation, based on a logistic regression model accounting for both sexual orientation and race as predictors of reported workplace discrimination. Here, non-heterosexual indi-

Table 3: Logistic Regression: Workplace Discrimination on Sexual Orientation with Controls

	(1)
(Intercept)	−0.510* (0.259)
sexorientNon-Heterosexual	0.277 (0.368)
raceOther	−0.373 (0.277)
raceWhite	−0.797*** (0.180)
age	−0.007 (0.005)
genderMale	0.044 (0.147)
Num.Obs.	1189
AIC	1179.8
BIC	1210.3
Log.Lik.	−583.907
RMSE	0.40

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

viduals remain more likely to report experiencing discrimination compared to heterosexual individuals, controlling for race. However, the confidence intervals for non-heterosexual respondents are substantially wider, suggesting greater uncertainty in the estimate, likely due to smaller sample size. Again, the overlap in confidence intervals indicates that this difference may not be statistically significant.

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[1] 0.8202247
```

Area under the curve: 0.5385

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# A tibble: 2 x 4
  sexorient      mean_age prop_male prop_white
  <fct>         <dbl>    <dbl>    <dbl>
1 Heterosexual    47.5    0.498    0.750
2 Non-Heterosexual 38.9    0.333    0.738
```

Preliminary Pew Research Center Data Visuals

Table 4: Does LGBT identity predict workplace discrimination

	(1)
(Intercept)	−1.059 (1.263)
factor(LGBT)2	0.352 (0.603)
factor(LGBT)3	−0.282 (0.367)
factor(LGBT)4	0.001 (0.772)
SEX	−0.030 (0.516)
ppagecat	0.066 (0.093)
PPETHM	0.169 (0.103)
PPEDUCAT	−0.081 (0.174)
INCOME	−0.126* (0.060)
Num.Obs.	761
R2	0.030
R2 Adj.	0.020
AIC	808.9
BIC	840.0
Log.Lik.	−386.827
RMSE	0.40

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001